Local and Latent: Dual Views of Symmetry Breaking in Cooperative Multi-Robot RL

PI: Qi Zhang, Assistant Professor, Computer Science Department, Worcester Polytechnic Institute

Cash funding needed: $42,450; **AWS Promotional Credits needed:** $50,000

Amazon contacts: Zhenyu Zhang ⟨zhenyuzh@amazon.com⟩, Özalp Özer ⟨ozalpo@amazon.com⟩

Abstract Spatial symmetry is one of the most effective inductive biases for learning in robotics, but in real multi-robot deployments it almost never holds exactly. We argue that the failures come in two structurally distinct forms — local breakers, visible in the current state but confined to small regions (e.g., fixed obstacles), and latent breakers, hidden from observation but fixed across an episode (e.g., wind or gravity acting on the whole team). The two are not interchangeable: state-dependent gating cannot find a breaker it does not observe, and hidden-context inference attributes all violations to a single global variable. We propose a dual-view theory and a unified multi-robot architecture that address both. View 1 formalizes a locally factored partially symmetric Markov game and shows that the critic-level symmetry error at a safe pair is controlled by its discounted future breaker exposure, motivating a TD-learned per-agent gated critic. View 2 formalizes a latent symmetry-breaker POMDP, bounds value error by breaker-estimation error, and establishes posterior equivariance — constraining the encoder design. The unified architecture pairs an equivariant latent encoder with per-agent state-dependent gating, and is evaluated on multi-robot tasks isolating each view and combining them.

Keywords cooperative multi-robot RL, group equivariance, symmetry breaking

1 Introduction

Group symmetries, geometric transformations that leave certain properties of a system invariant, provide one of the most effective inductive biases in robot learning. This is because the physical world is rich in geometric structure: a navigation controller should not depend on the absolute orientation of the world frame, and a pick-and-place skill executed at one workspace location should transfer to a rotated or translated configuration of the same objects. Encoding such structure directly into the learner reduces the effective hypothesis space and enables generalization from few samples, which has produced consistent gains in both imitation learning-based (e.g., [1, 2, 3]) and reinforcement learning (RL)-based (e.g., [4, 5] and PI's [6] by PI) robot learning, where exploration is expensive and high-quality demonstrations are scarce. To provide formality and prepare for theoretical grounding, we formalize symmetries in robot learning with the notion of group-symmetric Markov decision processes (MDPs) [4, 5]: Given a group G acting on the state space $\mathcal{S}$ and action space $\mathcal{A}$ of an MDP, the MDP is G-symmetric if the transition dynamics and reward function are invariant under the group actions, i.e., for every $g \in G$ and $(s, a, s') \in \mathcal{S} \times \mathcal{A} \times \mathcal{S}$,

$$P(g \cdot s' | g \cdot s, g \cdot a) = P(s' | s, a), \quad R(g \cdot s, g \cdot a) = R(s, a) \tag{1}$$

where $g\cdot$ denotes the group action on the state and action spaces.

Although orginally developed for single-robot settings, the notion of group-symmetric MDPs can be extended to multi-robot settings by extending the group actions and invariance conditions in (1) to the joint state, action, and observation spaces of all agents (PI's [7]). In multi-robot coordination, the relevant group typically factors as $G = G_{\text{spatial}} \times \mathcal{S}_n$, where G_{spatial} captures Euclidean symmetries (i.e., translations and rotations) and $\mathcal{S}_n$ captures permutation symmetry over n homogeneous agents. Commensurate with the time and resource constraints of this proposal, we will focus on the spatial symmetries in multi-robot coordination and leave permutation symmetry for future work.

In practice, however, the symmetry condition in (1) almost never holds exactly and globally in real-world multi-robot tasks. We observe that essentially every realistic violation falls into one of two structurally distinct modes, separated by a single diagnostic question: is the symmetry-breaking signal present in an agent's current observation, or hidden from it? This *local-latent* dichotomy is the organizing principle of this proposal, and the two modes call for fundamentally different mechanisms:

Local symmetry breaking occurs when the symmetry-breaking signal is present in an agent's current observation. This often happens in workspaces that contain fixed obstacles, walls, charging docks, asymmetric goal regions, etc. For example, for a team of warehouse robots navigating around a fixed shelving unit, 2D Euclidean symmetry is broken only in the immediate vicinity of the obstacle, whereas far from it, rotations and translations of the joint configuration leave the optimal team behavior invariant. Crucially, in a typical configuration only a *subset* of the agents is actually near the obstacle at any given moment; the rest continue to operate in regions where rotational symmetry is exact. The breaker here is observable in the current local state and localized to a small region of the per-agent state-action space.

Latent symmetry breaking, on the other hand, occurs when the symmetry-breaking signal is hidden from the agent's current observation and can potentially be inferred from historical information. In this proposal, we focus on the common scenario where the symmetry-breaking signal is a latent global variable that is fixed within an episode but varies across episodes. As examples, consider a team of aerial drones operating in a constant wind field, a multi-arm setup mounted on an inclined surface, and a fleet of underwater robots in a steady current, where all share the same structural property: the underlying dynamics would be *exactly symmetric if* the wind, gravity, or current vector were rotated together with the state and action, yey these vectors are not directly observable. Such symmetry breakers are neither localized in the visible state nor detectable from a single transition; they are latent variables that must be inferred from trajectory history.

The two modes are structurally orthogonal and not interchangeable. A method for local symmetry breaking has nothing to offer when the breaker is hidden, whereas a latent-inference method, conversely, attributes all symmetry violations to a single global variable and cannot efficiently represent localized violations whose effects depend on which agent is nearby. Realistic multi-robot deployments almost always exhibit both modes simultaneously, and we aim for a complete treatment that addresses both.

The focus of this proposal on the local-latent dichotomy separates it from prior works, which have either focused on single-agent settings where the dichotomy does not arise, or have taken a monolithic approach that globally relaxes the symmetry condition without distinguishing between structurally different modes of symmetry breaking (e.g., [8, 9, 10, 11, 12, 13]). Closing this gap is critical for the development of effective multi-robot learning algorithms and is the goal of this proposal.

2 Methods

We pursue two complementary views of spatial symmetry-breaking in multi-robot systems. *View 1* (§ 2.2) develops the multi-agent theory of agent-wise local breakers—when symmetry holds piecewise in each local factor—and yields a gated critic that routes equivariant and raw branches based on a learned state-dependent signal. *View 2* (§ 2.3) treats global hidden context (wind, gravity, current) as a latent symmetry breaker and recovers equivariance in the *augmented* state through breaker inference. The two views are not mutually exclusive: they correspond to different sources of symmetry violation and can be combined within a single agent architecture (§ 2.4).

Setting

We consider a discounted cooperative Markov game $\mathcal{M} = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$ with agent set $\mathcal{N} = \{1, \ldots, N\}$ and joint action space $\mathcal{A} = \mathcal{A}^1 \times \cdots \times \mathcal{A}^N$. Both views assume a spatial group G acting on the workspace, lifted to each agent's state and action via representations $(L_g^i, K_g^i) \colon \mathcal{X}^i \times \mathcal{A}^i \to \mathcal{X}^i \times \mathcal{A}^i$. The two views differ in *why* the symmetry G may fail to be exact: View 1 attributes failures to localized regions in the observed local state–action space; View 2 attributes failures to an unobserved global breaker variable.

View 1 — Agent-wise local routing for partially symmetric Markov games

Formal model

We posit a *locally factored partially symmetric Markov game* in which the joint state, reward, and transition decompose agent-wise:

$$\mathcal{S} = \mathcal{X}^1 \times \cdots \times \mathcal{X}^N, \qquad r(s,a) = \sum_{i=1}^{N} r_i(x^i, a^i), \qquad P(\cdot \mid s, a) = \bigotimes_{i=1}^{N} P_i(\cdot \mid x^i, a^i). \tag{2}$$

Each local factor carries its own finite symmetry group G_i acting on $(\mathcal{X}^i, \mathcal{A}^i)$ and its own *local breaker region* $\mathcal{B}^i \subseteq \mathcal{X}^i \times \mathcal{A}^i$. On the safe set $(x^i, a^i) \notin \mathcal{B}^i$ the local reward and transition are G_i-invariant; on $\mathcal{B}^i$ they may be arbitrary. The breaker region is itself G_i-invariant, so the safe/broken designation is a property of the orbit, not of a representative element.

This formalism is more general than a single global gate over the joint state. It allows symmetry to be exploited at agents whose local context is symmetric, even when other agents' local contexts are not—recovering the multi-robot nuance flagged in § 1.3(a).

Theoretical claims

We pursue two theoretical results that, together, justify a gated critic architecture.

Claim T1 (one-step model exactness). Define a routed model that, on each factor, uses the group-averaged reward and transition on the safe set and the raw local reward and transition on the breaker region:

$$\begin{aligned} \widetilde{r}_i(x^i, a^i) &:= (1 - \lambda_{\text{cur}}^i)\, \overline{r}_i(x^i, a^i) + \lambda_{\text{cur}}^i\, r_i(x^i, a^i), \\ \widetilde{P}_i &:= (1 - \lambda_{\text{cur}}^i)\, \overline{P}_i + \lambda_{\text{cur}}^i\, P_i, \end{aligned} \tag{3}$$

with $\lambda_{\text{cur}}^i(x^i, a^i) := \mathbf{1}\{(x^i, a^i) \in \mathcal{B}^i\}$ and $\overline{r}_i, \overline{P}_i$ the G_i-orbit averages. We will show that the routed Bellman operator $\widetilde{\mathcal{T}}$ coincides with the true $\mathcal{T}$, so $\widetilde{Q}^* = Q^*$. This establishes that oracle agent-wise local routing is *exact at the one-step model level* and isolates where the difficulty actually lies.

Claim T2 (future breaker exposure controls critic symmetry error). One-step model symmetry does *not* imply critic symmetry, because the critic sums discounted future rewards along trajectories that may eventually enter breaker regions. We will prove that for a safe-equivariant local policy π_i and any safe pair $(x, a) \notin \mathcal{B}^i$,

$$\left| Q_i^{\pi_i}(L_g^i[x], K_g^i[a]) - Q_i^{\pi_i}(x, a) \right| \;\le\; 2\, V_{i,\max}\, \beta_i^{\pi_i}(x, a), \tag{4}$$

where $\beta_i^{\pi_i}(x, a) := \mathbb{E}_{x,a}^{\pi_i}[\gamma^{\tau_i}]$ is the *discounted future breaker exposure* and τ_i is the first future hitting time of $\mathcal{B}^i$. A finite-horizon variant replaces β with $\eta_{i,H}^{\pi_i}(x, a) + \gamma^{H+1}$. The implication is that symmetry bias on the critic is *low-error* exactly where future breaker exposure is small.

Algorithm: TD-learned gated critic

Equation (4) motivates gating at the *critic level* rather than at the representation level: the value-function statement is most directly implemented by mixing two critic outputs. Concretely, we instantiate

- an *orbit-averaged invariant critic* $Q_\theta^{\text{inv}}(s,a) = \frac{1}{|G|}\sum_{g\in G} f_\theta(g\cdot s, g\cdot a)$,
- an unconstrained *raw critic* $Q_\phi^{\text{raw}}(s,a)$, and
- a learned gate $\widehat{\lambda}_\psi(s,a) = \sigma(g_\psi(s,a)) \in [0,1]$,

combined as $Q_{\theta,\phi,\psi}^{\text{gate}}(s,a) = (1-\widehat{\lambda}_\psi)\, Q_\theta^{\text{inv}} + \widehat{\lambda}_\psi\, Q_\phi^{\text{raw}}$. A *local-factor variant* sums per-agent gated branches and matches the hypotheses of Claim T2 more directly.

Critically, the gate is trained *purely* through the TD loss—no auxiliary supervision on breaker labels. The gradient $\partial Q^{\text{gate}}/\partial\widehat{\lambda}_\psi = Q_\phi^{\text{raw}} - Q_\theta^{\text{inv}}$ provides the routing signal: when the raw branch reduces TD error more than the invariant branch at (s,a), the gate opens; when the two branches agree, the gate is free to favor the more data-efficient invariant branch. We implement this within a CTDE actor–critic with decentralized actors $a^i = \mu_i(o^i)$ and a centralized gated critic.

Baselines and ablations

The empirical evaluation includes (i) a raw MLP critic, (ii) a fully invariant critic, (iii) a *current-breaker oracle gate* using ground-truth $\mathbf{1}\{(s,a) \in \mathcal{B}\}$—a diagnostic that tests whether *current* breaker information alone is sufficient—and (iv) the TD-learned gated critic (the main method). Per the future-exposure theory, we expect (iii) to be insufficient on tasks where trajectories cross breaker regions several steps into the future, while (iv) recovers the right gating signal through TD bootstrapping.

View 2 — Symmetry breakers as partially observed latent state

A note on notation. Unlike View 1, the formalism in this subsection is written in single-agent-style notation (s,a,o,b) rather than the agent-wise factorization $s = (x^1,\dots,x^N)$. This is deliberate: the symmetry-breaking mechanism studied here is *global* rather than agent-wise. A hidden wind or gravity vector is a single latent variable shared by *all* agents in the team and fixed for the entire episode, so the relevant structural object—augmented equivariance under simultaneous rotation of the state, action, and breaker—does not decompose factor-by-factor in the way View 1's local breaker regions do. Throughout § 2.3 we therefore use (s,a,o) to denote the joint state, joint action, and joint observation of the team, with the understanding that each is composed of per-agent quantities $s = (x^1,\dots,x^N)$, $a = (a^1,\dots,a^N)$, $o = (o^1,\dots,o^N)$, and that the spatial group G acts simultaneously on all agents' local frames. The breaker b remains a single global object. The agent-wise factorization re-enters explicitly in § 2.4, where we combine View 1 and View 2 in a properly multi-agent architecture.

Formal model

For hidden-context breakers (§ 1.3(b)), the agent-wise local-region view does not fit: the breaker is fixed throughout the episode and never appears in the observation. We instead model the environment as a *latent symmetry-breaker POMDP* $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{B}, P_b, r_b, \Omega_b, \gamma, \mu_0, \rho_0\rangle$ in which a hidden breaker $b \sim \rho_0$ is sampled at the start of each episode and held fixed. The agent receives observations $o_t \sim \Omega_b(\cdot \mid s_t)$ and history $h_t = (o_0, a_0, r_0, \dots, o_t)$ but never sees b.

The structural assumption is *augmented equivariance*: for every $g \in G$,

$$r_{g\cdot b}(g\cdot s, g\cdot a) = r_b(s,a), \quad P_{g\cdot b}(g\cdot C \mid g\cdot s, g\cdot a) = P_b(C \mid s,a), \quad \Omega_{g\cdot b}(g\cdot D \mid g\cdot s) = \Omega_b(D \mid s). \tag{5}$$

The system is symmetric *if and only if* b is transformed alongside (s,a). Holding b fixed while

rotating (s, a) therefore yields an incorrect comparison, and the visible state alone admits no group action consistent with the dynamics.

Theoretical claims

Claim T3 (no observation-only equivariance). If the dynamics are non-trivial in b, then any critic that enforces $Q(g \cdot o, g \cdot a) = Q(o, a)$ on the observation alone is inconsistent with the true Bellman targets at some (o, a, g). The correct invariance is $Q(g \cdot o, g \cdot a, g \cdot z) = Q(o, a, z)$ where z is a breaker-sufficient statistic of history.

Claim T4 (value sensitivity to breaker-estimation error). Under Lipschitz assumptions $|r_b - r_{b'}| \leq L_r d_{\mathcal{B}}(b, b')$ and $\|P_b - P_{b'}\|_{\mathrm{TV}} \leq L_p d_{\mathcal{B}}(b, b')$, conditioning on an estimate $\widehat{b}$ instead of the true b yields a fixed-point value error bounded by $\frac{L_r + \gamma V_{\max} L_p}{1-\gamma} d_{\mathcal{B}}(b, \widehat{b})$. This quantifies the benefit of better breaker inference and motivates auxiliary objectives that drive $d_{\mathcal{B}}(b, \widehat{b})$ down.

Claim T5 (equivariant belief update). Under (5) and an equivariant prior ρ_0, the Bayesian posterior over b transforms equivariantly: $\beta_t^{g \cdot h_t}(g \cdot B) = \beta_t^{h_t}(B)$. The encoder f_η should therefore be built so that its output transforms in the same representation as b (e.g., a 2D vector for a 2D wind direction).

Algorithm: breaker-conditioned actor–critic

The architecture has three components:

1. *Latent breaker encoder* $z_t = f_\eta(h_t)$, implemented as a recurrent network or a contrastive embedding. The encoder is designed so that $f_\eta(g \cdot h_t) = g \cdot f_\eta(h_t)$—i.e., its output lives in the same representation as b.
2. *Breaker-conditioned policy and critic*, equivariant in the augmented pair (o, z):

$$\pi_\theta(g \cdot a \mid g \cdot o, g \cdot z) = \pi_\theta(a \mid o, z), \qquad Q_\omega(g \cdot o, g \cdot a, g \cdot z) = Q_\omega(o, a, z).$$

3. *Auxiliary objectives* for the encoder, used in addition to the TD signal: a dynamics prediction loss $\mathcal{L}_{\mathrm{dyn}} = \mathbb{E}[\ell(D_\psi(o_t, a_t, z_t), o_{t+1})]$, a contrastive loss that pulls together representations from the same episode (same b) and pushes apart those from different episodes, and an optional supervised loss $\mathcal{L}_{\mathrm{sup}} = \mathbb{E}[d_{\mathcal{B}}(f_\eta(h_t), b)^2]$ when simulator labels are available.

We will explore three implementation variants: a *point-estimate* encoder producing $\widehat{b}_t \in \mathcal{B}$; a *belief* representation outputting distributional parameters; and *canonicalization*, in which the network rotates inputs into a frame aligned with $\widehat{b}_t$, applies a standard network, and rotates outputs back—turning latent inference into a learned coordinate gauge.

Baselines and ablations

We compare against (i) an MLP actor–critic with no symmetry prior; (ii) a strict observation-only equivariant model; (iii) an *oracle* breaker-conditioned equivariant model that receives the true b, providing an upper bound; (iv) a recurrent non-equivariant model that has access to history but no symmetry prior; and (v) an *observable-breaker ablation* that exposes b in the observation, confirming that the latent variable is indeed the missing information.

Unifying perspective and combined architecture

Although developed independently, the two views compose naturally because they operate at *different granularities*. View 1 introduces a *per-agent, state-dependent* mechanism: each local factor (x^i, a^i) has its own breaker region $\mathcal{B}^i$ and its own gate $\widehat{\lambda}^i$. View 2 introduces a *global, per-episode* mechanism: a single latent breaker $b \in \mathcal{B}$ is shared by the whole team and inferred once from the joint trajectory history. The two mechanisms address orthogonal failure modes—localized obstacle

interactions on one hand, hidden physical context on the other—and can be composed without conflict.

Multi-agent setting. As in § 2.2, the joint state and action factor agent-wise: $s = (x^1, \ldots, x^N)$ and $a = (a^1, \ldots, a^N)$. As in § 2.3, the team operates under partial observability with joint observation $o_t = (o_t^1, \ldots, o_t^N)$ and joint history $h_t = (o_0, a_0, r_0, \ldots, o_t)$. A single hidden breaker $b \sim \rho_0$ is sampled per episode and remains fixed; each agent i also has its own local symmetry group G_i and local breaker region $\mathcal{B}^i \subseteq \mathcal{X}^i \times \mathcal{A}^i$.

Global breaker inference. A shared encoder $z_t = f_\eta(h_t)$ ingests the *joint* history and produces a global breaker estimate that is broadcast to every agent. The encoder is built to satisfy the augmented-equivariance condition $f_\eta(g \cdot h_t) = g \cdot f_\eta(h_t)$—when the entire team's history is rotated, the inferred latent rotates with it. In a CTDE implementation, f_η runs in the centralized critic and conditions all per-agent branches.

Per-agent gated, breaker-conditioned critic. For each agent i, define a *breaker-conditioned invariant branch*

$$q^{\text{inv}}_{i,\theta}(x^i, a^i, z_t) \;=\; \frac{1}{|G_i|} \sum_{g \in G_i} f_{i,\theta}\big(L^i_g[x^i],\, K^i_g[a^i],\, g \cdot z_t\big), \tag{6}$$

which is equivariant in the *augmented* per-agent triple (x^i, a^i, z_t)— the breaker estimate transforms together with the local state and action, following the augmented equivariance of View 2. Define a *per-agent raw branch* $q^{\text{raw}}_{i,\phi}(x^i, a^i, h_t)$ that has access to the joint history and is otherwise unconstrained. Mix them with a per-agent gate $\widehat{\lambda}^i_\psi(x^i, a^i) \in [0, 1]$ from View 1:

$$\begin{aligned} Q_i(x^i, a^i, h_t) = \big(1 - \widehat{\lambda}^i_\psi(x^i, a^i)\big)\, q^{\text{inv}}_{i,\theta}(x^i, a^i, z_t) \\ +\; \widehat{\lambda}^i_\psi(x^i, a^i)\, q^{\text{raw}}_{i,\phi}(x^i, a^i, h_t). \end{aligned} \tag{7}$$

The total centralized critic is the agent-wise sum

$$Q(s, a, h_t) \;=\; \sum_{i=1}^{N} Q_i(x^i, a^i, h_t), \tag{8}$$

matching the factored value structure of (2). This factorization is what makes the local theoretical bound of Claim T2 directly applicable to the combined model: each summand individually satisfies the per-agent future-exposure bound, conditioned on z_t.

Reading the two mechanisms. The invariant branch (6) is the one that carries the inductive bias from both views simultaneously. It exploits *local* G_i-equivariance in agent i's frame, but only insofar as the global hidden context has been correctly inferred—captured by conditioning on z_t. The per-agent gate $\widehat{\lambda}^i_\psi$ allows agent i to fall back to its raw branch when its *local* state–action pair lies in $\mathcal{B}^i$, regardless of how well the global breaker has been estimated. Conversely, a poor estimate z_t does not corrupt agents whose gates are already open—those agents already route through $q^{\text{raw}}_{i,\phi}$. The two mechanisms therefore degrade gracefully and independently.

Training. The TD loss on the joint critic

$$L_Q = \mathbb{E}_{\mathcal{D}}\big[(Q(s_t, a_t, h_t) - y_t)^2\big]$$

with $y_t = r_t + \gamma\, \overline{Q}(s_{t+1}, a'_{t+1}, h_{t+1})$ propagates gradients to all three parameter groups: the equivariant branches $\{\theta\}$ and raw branches $\{\phi\}$, the per-agent gates $\{\psi\}$, and the encoder η. The

encoder additionally receives gradients from the auxiliary objectives of § 2.3.3 (dynamics, contrastive, optional supervised). No gate-supervision loss is used; the routing decisions emerge from TD bootstrapping in the standard way.

Empirical claim. We will evaluate (7)–(8) against the single-view variants of § 2.2.3 and § 2.3.3, predicting that the combined architecture matches the better of the two single-view models on each task family and strictly dominates them on tasks that mix both failure modes (e.g., obstacles *and* hidden wind). Confirming this orthogonality is a primary empirical claim of this proposal.

Experimental design

We will evaluate across three task families, each chosen to isolate a different combination of the symmetry-breaking modes:

1. **Multi-robot navigation with fixed obstacles.** Cooperative goal-reaching on a 2D arena with a small number of fixed asymmetric obstacles. This isolates *agent-wise local breakers* (View 1); a subset of agents at any given timestep is near an obstacle while others are not. Symmetry group $G = \mathrm{SO}(2)$ or a discrete subgroup. Symmetry-breaking strength is swept by varying obstacle density.
2. **Hidden wind navigation.** Same as task 1 but with an obstacle-free arena and a per-episode constant wind vector omitted from observations. This isolates *hidden global breakers* (View 2). The wind direction must be inferred from the discrepancy between commanded and realized displacement.
3. **Combined regime.** A multi-arm manipulation task on an inclined surface (hidden gravity direction) with localized obstacles. This tests the combined architecture (§ 2.4) and the orthogonality claim.

Metrics follow standard practice—final return, sample efficiency, training stability across seeds—supplemented by diagnostics specific to each view: gate activation maps and their correlation with oracle breaker labels (View 1) and angular error between $\widehat{b}_t$ and the true wind/gravity vector (View 2). Performance will be reported as a function of symmetry-breaking strength (obstacle density, wind magnitude) to characterize the regime where each view, and the combined architecture, dominates.

3 Expected Results

The table below shows a timeline for the key milestones of the project.

	M1	2	3	4	5	6	7	8	9	10	11	12	13	14
Setup	Baselines													
View 1		Theory			Algorithms									
View 2		Theory			Algorithms									
Unification									Views 1+2					
Code & Paper													Submission	

4 Funds Needed

Cash funding. Per PI's institution policy [14], $42,450 is requested for the minimum annual stipend amount for a research assistant (RA).

AWS promotional credits. We request $50,000 in AWS promotional credits over a one-year period.

The majority (∼$35,000) will be allocated to Amazon EC2 (P4d or G5 GPU instances alongside high-CPU instances) that will power parallelized (multi-robot) RL training. Approximately $10,000 will be directed toward AWS RoboMaker and Amazon SageMaker to enable realistic and large-scale robotics simulation, training, and benchmarking. The remaining ∼$5,000 will cover supporting infrastructure costs such as data storage and transfer.

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